

CLINICAL INVESTIGATION

Early Use of a Risk-Adjusted Mechanical Ventilation Digital Quality Measure Bundle in a Large Health System

OBJECTIVES: To describe the development, validation, and deployment of a risk-adjusted digital quality measure (dQM) bundle for spontaneous awakening trials (SATs), spontaneous breathing trials (SBTs), and low-tidal volume ventilation (LTVV) as part of a quality improvement (QI) program in a large health system.

DESIGN: Quasi-experimental before-after study.

SETTING: Thirty-seven ICUs across 14 hospitals in the United States.

PATIENTS: Mechanically ventilated patients older than 16 years.

INTERVENTIONS: An available, open-source, hospital mortality model, a new gradient-boosted ICU mortality model, and four new, heterogeneous, stacked ensemble predicted duration of mechanical ventilation (DMV) models (one model predicting up to 14 d of ventilation [14-d DMV model] and three multiple classifier models predicting up to 6 d of ventilation) were created. A regularly refreshing dashboard displaying risk-adjusted information was coupled with audit and feedback sessions for ICU leadership beginning in September 2020.

MEASUREMENTS AND MAIN RESULTS: Risk model performance was evaluated, as appropriate, with C-statistics, mean SE (MSE), concordance correlation coefficients (CCCs), and F1-scores. Across all ICUs, compliance with SBTs improved from 81 to 97%, LTVV 80 to 90%, and SATs 27 to 65%. Both hospital and ICU mortality models had robust performance, with C-statistics of 0.85 (95% CI, 0.84–0.85) and 0.94 (0.93–0.94), respectively. The 14-day DMV model MSE was 0.63 and CCC was 0.97, whereas the multiple classifier DMV models F1-scores ranged from 0.42 to 0.59. Unadjusted DMV was greater post-implementation (4.32 ± 3.99 d) vs. pre-implementation (3.76 ± 3.66 d). Actual vs. predicted ventilator days were stable pre-implementation vs. post-implementation when assessed with the multiple classifier models and decreased in the post-implementation period when assessed with the 14-day model. Risk-adjusted mortality remained stable.

CONCLUSIONS: A dQM bundle proved useful for efficiently tracking process measures related to a ventilator management QI program in a large health system, although risk-adjusted information differed depending on model constructs. Future work should focus on developing and validating generalizable and interoperable dQM bundles.

KEYWORDS: electronic health record; learning health system

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Several proven, evidence-based care components constitute best practices in the treatment of critically ill adults who require mechanical ventilation. Spontaneous awakening trials (SATs), spontaneous breathing trials (SBTs), and low-tidal volume ventilation (LTVV) are associated with improved patient outcomes, including reduced time to extubation (SBT) and

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KEY POINTS

Question: Can a digital quality measure (dQM) bundle help track and improve adherence to best-practice mechanical ventilation strategies in ICUs?

Findings: Implementation of a ventilator management dQM bundle as part of an audit and feedback strategy across 37 ICUs was associated with improved compliance with spontaneous awakening trials (27–65%), spontaneous breathing trials (81–97%), and low-tidal volume ventilation (80–90%), with stable to possibly improved risk-adjusted ventilator days.

Meanings: A risk-adjusted dQM ventilator bundle helped to efficiently promote adherence to evidence-based ventilator practices and highlight the potential for scalable, interoperable analytics in large health systems.

lower mortality (SAT, LTVV) (1–3). As a result, these elements have been incorporated into care bundles and guidelines endorsed by professional societies and organizations, including the Society of Critical Care Medicine (4), the American Thoracic Society (5), and the Agency for Healthcare Research and Quality (6).

As with most evidence-based practices, promoting adherence to ventilator management bundles requires well-constructed quality improvement (QI) programs to ensure adoption and ongoing compliance. Commonly used QI paradigms, including the Institute for Healthcare Improvement's (IHI) Model for Improvement (7), Kotter's change management model (8), Lewin's change model (9, 10), Lean (11), Six Sigma (10), and Shewhart cycles (12), all rely on iterative review of relevant, validated metrics. Real-world data also form the foundation of the National Academy of Medicine's learning health system paradigm (13). Accordingly, well-executed data collection, curation, and reporting are essential for successful continuous QI. However, high-quality transformation of data to knowledge is time-intensive and costly. At one hospital system, preparation of quality metrics amounted to 108,478 person-hours, greater than \$5 million in personnel costs, and greater than \$600,000 in vendor fees annually. In contrast, electronic metrics were noted to consume far less resources at \$1901.58 per metric per year (14).

Digital quality measures (dQMs) are a method of electronically tracking system-level performance that has been put forward by the Centers for Medicare & Medicaid Services (CMS) as a means for health system and hospital benchmarking at substantially reduced costs compared with manual data curation (15). CMS has developed a strategic roadmap for promoting the adoption of dQMs, much of which depends on the ongoing implementation of mandated interoperability standards by electronic health record (EHR) vendors (16). In advance of interoperable dQMs being widely adopted and available to promote best ventilator practices across ICUs, early development of scalable analytics may help QI teams promote standardized, evidence-based, best practices (17). This approach may be especially efficient in large health systems and for data-intensive arenas like critical care medicine, for which the burden of manual data curation is greatest (18). In support of this theory, we describe the development and deployment of dQMs for SATs, SBTs, and LTVV in critically ill patients as part of a QI program at a large health system focused on reducing unnecessary days of mechanical ventilation and explore possible associations with mechanical ventilation days using novel, validated, open-source risk adjustment models.

METHODS

Study Design, Organization Description, and Regulatory Approval

This is a prospective, observational, cohort study that presents the results of one aspect of an ongoing, multi-year QI effort across ICUs in UPMC, a large health system in the United States. UPMC is an integrated delivery and finance system with 40 community, tertiary, and quaternary hospitals in 3 states. The UPMC ICU Service Center is responsible for financial stewardship initiatives, guideline development, and workflow standardization across 45 adult and PICUs within the health system. Additional details of the ICU Service Center structure are in the **Supplemental Model Methods** (<https://links.lww.com/CCM/H734>). This work was approved by UPMC QI leadership in January 2020, the subsequently reviewed and approved by the UPMC QI review committee, with final approval for publication of the findings granted on April 4, 2025. Projects reviewed and approved by the UPMC

QI review committee do not meet the federal definition of research according to 45 CFR 46.102(l) and do not require additional institutional review board (IRB) oversight; however, the University of Pittsburgh IRB approved waiver of informed consent on November 1, 2023, for the portions of the work requiring machine learning classification modeling by Dr. Li. Accordingly, this work adhered to both institutional ethics standards and the Helsinki Declaration of 1975. Findings for the QI initiative are reported according to the Standards for Quality Improvement Reporting Excellence 2.0 (**Supplemental Table 1**, <https://links.lww.com/CCM/H734>) (19).

Ventilator Quality Improvement Program Development

The purpose of the ventilator QI program was to reduce unnecessary days of mechanical ventilation, as measured by the difference between actual and predicted ventilator days. Metric selection was guided by the best practices outlined in the Society of Critical Care Medicine's ICU Liberation Bundle (also known as the A-F bundle) (4), which consists of six components: 1) Assess, prevent and manage pain, 2) perform Both SATs and SBTs, 3) Choose the right analgesia and sedation regimen, 4) Delirium assessment, management, and prevention, 5) Early mobility and exercise, and 6) Family engagement and empowerment. The ventilator QI program focused on component B. SATs are tracked using programed logic that accounts for the presence of expected sedative infusion, mechanical ventilation, and the presence or absence of contraindications to SAT performance. SBTs are tracked for patients receiving two or more consecutive days of mechanical ventilation and account for the presence or absence of documented contraindications to SBTs. Although not part of the A-F bundle, LTVV was tracked as substantial evidence supports using lower ventilator tidal volumes as part of best practices for mechanical ventilation to reduce lung injury and mortality (2, 20). **Supplemental Table 2** (<https://links.lww.com/CCM/H734>) details the programming logic for the SAT, SBT, and LTVV metrics (ventilator dQM bundle).

Development of the ventilator dQM bundle and an associated audit and feedback approach to change management built upon related efforts that pre-dated the ICU Service Center's formation in 2017. This

included an SBT compliance program in 12 ICUs (21), and a separate program that accomplished improved LTVV compliance across 10 ICUs (22). The impact of these preceding efforts on patient-centered outcomes was uncertain. One program focused entirely on process measures and another program demonstrated inconsistent improvements inpatient outcome measures. Collectively, however, these programs afforded opportunities for post-implementation reviews that included several key lessons learned and areas for improvement that informed the design and implementation strategy of a more robust ventilator management bundle. The translation of lessons learned into the ventilator dQM bundle and associated change management approach is detailed in **Supplemental Table 3** (<https://links.lww.com/CCM/H734>), which summarizes the findings of previous post-implementation reviews. To support these initiatives, compliance with SATs and SBTs was integrated into an opening page of the EHR. Other components of the A-F bundle were also integrated into the EHR display and were the focus of previous QI initiatives. **Supplemental Table 4** (<https://links.lww.com/CCM/H734>) summarizes other UPMC QI initiatives focused on A-F bundle elements and provides additional context for the present work.

Approach to Practice Change Management

The ICU Service Center team holds monthly meetings with clinical leadership from all the ICUs in the health system to discuss leadership priorities and related progress on select metrics. Separate, twice annual, hybrid in-person and virtual meetings are held with each hospital's ICU clinical leadership and stakeholders to review key performance indicators for the hospitals' ICUs and compare performance to ICUs throughout the health system and to peer ICUs as designated by community, tertiary, or quaternary hospital status. Data are available to clinical leadership on-demand through a Qlik visualization dashboard that refreshes monthly (Qlik, King of Prussia, PA). Performance indicators are color-coded according to progress relative to system-set goals, with red indicating poor compliance (< 60% compliance with a metric), yellow indicating moderate compliance (60–90% compliance), and green indicating full compliance (> 90%). The ventilator dQM bundle was incorporated into a dashboard and audit and feedback sessions beginning

in September 2020, which marked the start of the “post-implementation” period.

Risk Adjustment Models Development, Validation, and Deployment

Risk-adjusted mortality was included as a balancing measure to detect unintended effects of the intervention and to account for potential changes in ventilator duration that might result from variations inpatient outcomes unrelated to the QI initiative. Tracking performance with the ventilator dQM bundle required accounting for varying patient illness severity across the 37 ICUs. To accomplish risk adjustment using electronic methods, an available, open-source mortality model, the “Global, Open-Source, Severity of Illness Score (GOSSIS),” was first adapted to an informatics pipeline in the system’s local warehouse architecture, yielding an “electronic GOSSIS” (eGOSSIS) presented alongside the dQM bundle in the visualization dashboard and feedback sessions (23). As there are no widely available, well-calibrated, open-source models for ICU mortality or predicted DMV, the implemented mortality model extraction, transform, and load process then served as the starting point for developing two more models for display in the dashboard and feedback sessions: a mortality model internally trained and calibrated on UPMC data, and a predicted duration of mechanical ventilation (DMV) model calculated using indicators of disease severity at the time of initiation of mechanical ventilation, as well as mortality and length of stay. The DMV model is a stacked, heterogeneous ensemble regressor trained to predict 1–14 days of mechanical ventilation. Finally, to improve upon limitations in the DMV model such as the risk of a recurrent relationship between predictors and the outcome, and to attempt to corroborate QI insights provided by the model for this report, three multiple classification models were developed to predict DMV up to 6 days, bucketing days of ventilation into four classes: 1, 2, 3, and 4–6 days of mechanical ventilation and avoiding potentially recurrent predictors. Additional details of the models’ development are in the **Supplemental Methods** (<https://links.lww.com/CCM/H734>) and are reported according to the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis for cluster analyses (**Supplemental Table 5**, <https://links.lww.com/CCM/H734>), with bias interrogation of models reported according to Prediction model Risk Of Bias Assessment Tool (**Supplemental Table 6**, <https://links.lww.com/CCM/H734>) (24, 25). **Supplemental Table 7**

(<https://links.lww.com/CCM/H734>) displays standard vocabulary mappings for model variables and **Supplemental Table 8** (<https://links.lww.com/CCM/H734>) displays curation logic for continuous measures. Standard vocabulary value sets for the eGOSSIS, UPMC Mortality Model, and UPMC predicted ventilator duration model were filed with the National Library of Medicine (26). Models were trained using data before September 2020.

Statistical Analysis

Continuous measures with a normal distribution are presented as means with 95% CIs and measures not normally distributed are presented as median with interquartile ranges (IQRs). Statistical performances of the mortality risk adjustment models were assessed using the area under the receiver operating characteristics curve (AUROC) and the area under the precision-recall curve (AUPRC). Calibration was assessed by viewing reliability diagrams and calculating Brier scores. Mortality model performance was benchmarked against a commercial mortality model (Vizient Inc, Irving, TX). The performance of the model for predicting the DMV was assessed by calculating the mean square error (MSE), root mean square error (RMSE), coefficient of determination (R^2), and the mean absolute error. The calibration of the duration model was assessed by viewing a reliability diagram and calculating the mean concordance correlation coefficient (CCC). Precision, recall, and F1-score were used to assess the performance of the multiple classifier model, along with heat maps and reliability plots. Opportunities and improvements were identified by plotting the actual and predicted average DMV as a run chart and identifying areas in which the ratio of actual to predicted duration of ventilation was $\pm 5\%$. Data curation was performed using Oracle SQL Workbench version 21.4.1.349 (Oracle Corp, Austin, TX), RStudio version 2023.09.1+494 “Desert Sunflower” (Posit, Boston, MA), R version 4.2.2 (R Foundation, Vienna, Austria), and Python version 3.9.13 (Python Software Foundation, Wilmington, DE). R and Python libraries are listed in the Supplemental Methods (<https://links.lww.com/CCM/H734>).

RESULTS

Patient Characteristics and Summary of Ventilator dQMs Across ICUs

Cohort ascertainment is described in **Supplemental Figures 1–3** (<https://links.lww.com/CCM/H734>).

Patient characteristics in the baseline and post-implementation periods are in **Table 1**. There were 29,380 patients in the baseline period and 41,787 in the post-implementation period. The majority were male, White race, and had a median age of 64 (IQR 53, 73). There was a greater proportion of deaths in the post-implementation period, a greater proportion of patients admitted to the ICU from the emergency department, and a higher severity of presenting illness, as indicated by a higher predicted mortality using the eGOSSIS. A greater proportion of patients were admitted to tertiary hospitals and fewer were admitted to community hospitals. **Supplemental Tables 9–11** (<https://links.lww.com/CCM/H734>) display the patient and predictor characteristics in the entire cohort, training cohort, and test cohort used for the UPMC Mortality Model and DMV models. **Table 2** displays SAT, SBT, and LTVV process measure compliance by 6-month epochs from 2020 to 2023, in aggregate and stratified by community, tertiary, and quaternary hospital type. The greatest improvements were observed in SAT compliance, which increased from 27% to 65% compliance overall. All hospital types demonstrated improvements in all process measures, with community hospitals demonstrating the greatest improvements across all measures and quaternary hospitals demonstrating the least improvements. At the end of 2023, all hospital types remained below the SAT compliance goal of 90%, and quaternary hospitals' LTVV compliance was 88%, below the 90% compliance goal.

Risk Model Performance

There were 146,039 encounters with an ICU admission for assessing the eGOSSIS performance. The AUROC was 0.85 (95% CI, 0.84–0.85) and the AUPRC was 0.55 (95% CI, 0.55–0.56), indicating excellent overall performance. **Supplemental Table 12** (<https://links.lww.com/CCM/H734>) details the statistical performance in all hospitals and stratified by community, tertiary, and quaternary hospital types. Calibration was poor, with a standardized mortality ratio (SMR) for all hospitals of 0.76 (95% CI, 0.75–0.77), ranging from 0.72 to 0.81 in hospital subtypes and a Brier score of 0.1 for all hospitals, ranging from 0.08 to 0.11 in hospital subtypes. In a subset of 500 patients at two quaternary hospitals using the Vizient predicted hospital mortality model, the eGOSSIS AUROC was 0.88 (95% CI, 0.84–0.93) and the

Vizient AUROC was 0.88 (95% CI, 0.83–0.92). In this cohort, the eGOSSIS SMR was 0.79 (95% CI, 0.63–0.96) and the Vizient SMR was 1.50 (95% CI, 1.19–1.82).

For the UPMC Mortality Model, there were 102,445 patients in the training cohort and 43,905 patients in the test cohort. **Supplemental Table 13** (<https://links.lww.com/CCM/H734>) displays the statistical performance characteristics across all hospitals and within the strata of hospital types in both the train and test cohorts. Performance was overall excellent, as indicated by an AUROC of 0.94 (95% CI, 0.93–0.94) and AUPRC 0.74 (0.73–0.75) in the test cohort. Calibration was excellent in the training cohort with an SMR of 1.0 (95% CI, 1.0–1.0) and a Brier score of 0.03 for all hospitals, and slightly worse in the test cohort with an SMR of 1.1 (1.0–1.1) and a Brier score of 0.05. **Supplemental Figures 4–11** (<https://links.lww.com/CCM/H734>) show the AUROC and calibration for the mortality models.

Supplemental Table 14 (<https://links.lww.com/CCM/H734>) displays the predicted DMV models' performances based on five-fold internal cross-validation. The 14-day DMV model, which included length of stay and mortality as features, demonstrated excellent performance and calibration, with an MSE of 0.63, RMSE of 0.8, MAE of 0.48, R^2 0.95, and CCC of 0.97. The three models developed using a multiple classification approach also demonstrated acceptable reliability. **Supplemental Table 15** (<https://links.lww.com/CCM/H734>) displays the performance characteristics of the three multiple classification DMV models. **Supplemental Figures 12–15** (<https://links.lww.com/CCM/H734>) display reliability diagrams for the DMV models. Heat maps of predictions for the multiple classifier models are displayed in **Supplemental Figures 16–18** (<https://links.lww.com/CCM/H734>).

Risk-Adjusted Performance

In the post-implementation period, total actual days of mechanical ventilation were less than the total predicted days of mechanical ventilation when assessed using all models. **Table 3** displays the actual and predicted mean DMV, as well as the total actual and predicted ventilator days, in the baseline and post-implementation periods. **Figure 1** displays actual and predicted mean DMV alongside the dQM measures and SMRs calculated with both the eGOSSIS and the UPMC Mortality Model,

TABLE 1.**Characteristics of Patients Receiving Mechanical Ventilation in an ICU During the Baseline Period and Post-Implementation of the Ventilator Quality Improvement Program**

Characteristic ^a	Baseline ^b	Post-Implementation ^c
N/n (%)	25,150	36,319
Median age, interquartile range (yr)	64 (53, 73)	64 (53, 73)
Sex		
Male	14,495 (58%)	21,179 (58%)
Female	10,655 (42%)	15,129 (42%)
Unknown		11 (< 1%)
Hospital type		
Community	1708 (7%)	2294 (6%)
Tertiary	6981 (28%)	18,911 (52%)
Quaternary	16,461 (66%)	15,114 (42%)
COVID-19 infection	137 (< 1%)	1987 (5%)
Predicted mortalities (%)	35	37
Actual mortality (%)	23	31
ICU admission source		
Emergency department	11,717 (47%)	19,249 (53%)
Operating room/recovery	4309 (17%)	4740 (13%)
Other hospital	9126 (36%)	12,239 (34%)
Other ICU	0 (0%)	91 (< 1%)
Primary diagnosis category ^d		
Cardiovascular	6993 (28%)	8443 (23%)
Gastrointestinal	3080 (12%)	3801 (10%)
Genitourinary	523 (2%)	718 (2%)
Gynecological	7 (< 1%)	6 (< 1%)
Hematological	310 (1%)	443 (1%)
Metabolic	544 (2%)	774 (2%)
Musculoskeletal/skin	490 (2%)	705 (2%)
Neurologic	2668 (11%)	4806 (13%)
Other medical disorders	577 (2%)	738 (2%)
Respiratory	4207 (17%)	4994 (14%)
Sepsis	4291 (17%)	8738 (24%)
Trauma	1460 (6%)	2155 (6%)

^aData are expressed as N/n (%) for categorical data, median (25th percentile, 75th percentile) for skewed data, and mean $\pm$ SD for normally distributed data.

^bThe duration of mechanical ventilation model was developed in a baseline (pre-implementation) cohort ascertained from January 1, 2018, to August 31, 2020, and performance was assessed using five-fold internal cross-validation.

^cPatient characteristics and related predictors for the duration of the mechanical ventilation model for the post-implementation period, September 1, 2020, to December 31, 2023, are displayed.

^dDiagnosis categories are per the Acute Physiology and Chronic Health Evaluation body system classification, which is used in model predictions.

^eAlaskan native, Hawaiian, Japanese, Samoan, and other.

^fGynecological and other.

TABLE 2.

Compliance With the Digital Quality Measure Bundle Over Time Across All Hospitals and Stratified by Community, Tertiary, and Quaternary Hospital Status

Year	2020 ^a	2021	2022	2023
Quarters (Q)	Q1-2/Q3-4 ^a	Q1-2/Q3-4	Q1-2/Q3-4	Q1-2/Q3-4
All hospitals compliance				
SAT	27%/28%	36%/33%	46%/49%	62%/65%
SBT	81%/86%	94%/94%	94%/94%	96%/97%
LTVV	80%/82%	85%/89%	86%/88%	89%/90%
Community hospitals compliance				
SAT	23%/31%	37%/36%	56%/69%	74%/78%
SBT	52%/74%	90%/93%	96%/94%	96%/97%
LTVV	79%/75%	78%/85%	83%/87%	85%/92%
Tertiary hospitals compliance				
SAT	18%/24%	32%/28%	42%/46%	54%/59%
SBT	75%/86%	94%/94%	94%/95%	96%/97%
LTVV	74%/78%	86%/87%	87%/87%	90%/93%
Quaternary hospitals compliance				
SAT	35%/31%	39%/39%	49%/49%	67%/70%
SBT	88%/87%	94%/94%	95%/94%	95%/98%
LTVV	83%/85%	86%/91%	86%/89%	89%/88%

LTVV = low-tidal volume ventilation, SAT = spontaneous awakening trial, SBT = spontaneous breathing trial.

^aPeriod during which the ventilator quality improvement program was implemented.

TABLE 3.

Actual and Predicted Days of Mechanical Ventilation in the Pre-Implementation and Post-Implementation Periods (14-d Model)

Hospital Type	N/n	Mean Duration of Mechanical Ventilation (SD)	Predicted Mean Duration of Mechanical Ventilation (SD)	Total Ventilator Days	Total Predicted Ventilator Days
Baseline January 1, 2018 to August 31, 2020					
All hospital types	25,150	3.76 (3.66)	3.76 (3.26)	94,611	94,701
Community	1,708	3.63 (3.19)	3.65 (2.81)	6,207	6,234
Tertiary	6,981	3.65 (3.51)	3.78 (3.17)	25,514	26,404
Quaternary	16,461	3.82 (3.77)	3.77 (3.35)	62,890	62,062
Post-Implementation September 1, 2020 to December 31, 2023					
All hospital types	36,319	4.32 (3.99)	4.39 (3.54)	156,864	159,470
Community	2,294	4.18 (3.70)	4.18 (3.27)	9,586	9,599
Tertiary	15,114	4.37 (3.98)	4.47 (3.53)	66,052	67,558
Quaternary	18,911	4.30 (4.03)	4.35 (3.58)	81,226	82,314

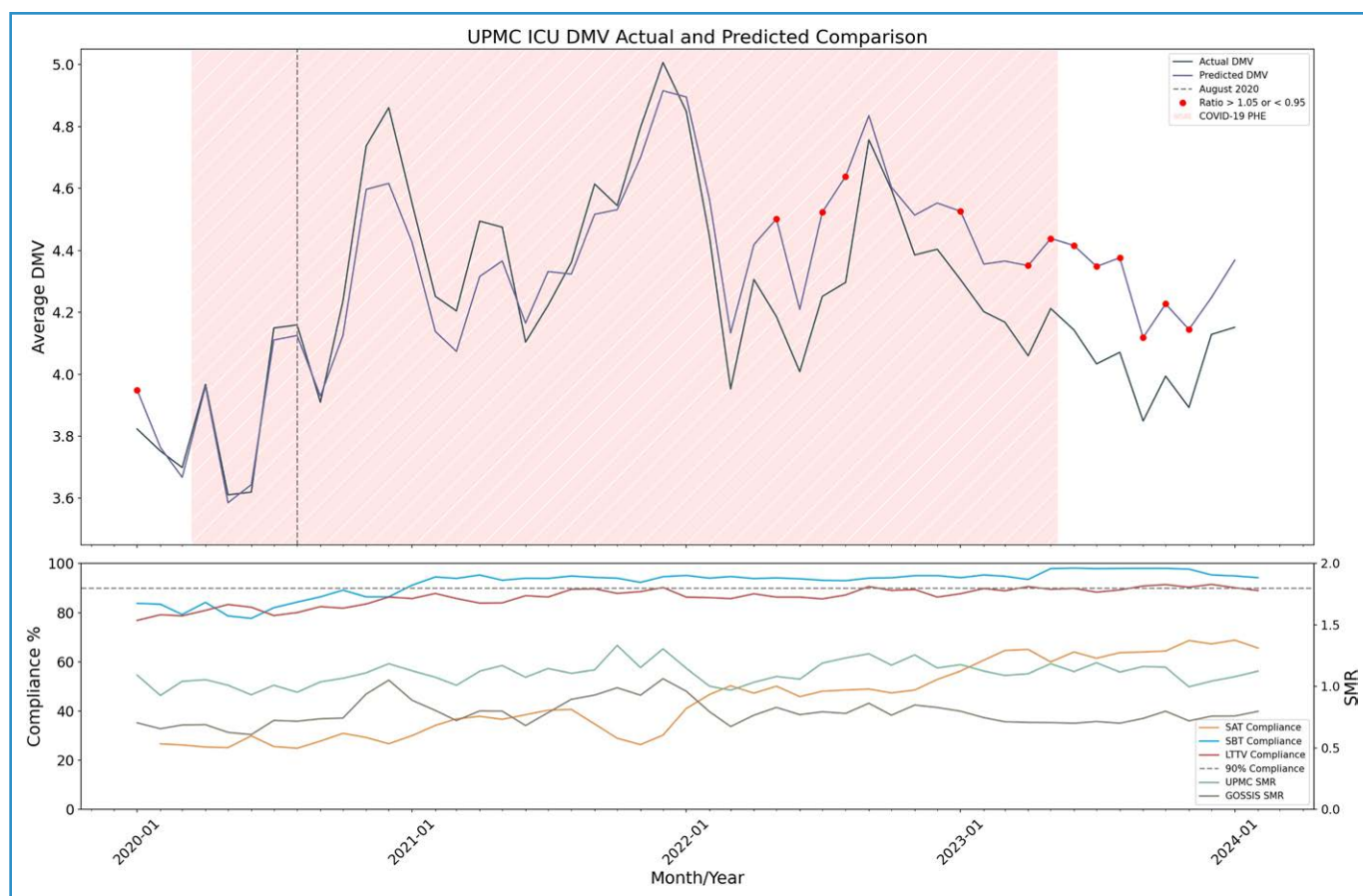


Figure 1. The *top plot* displays actual (black line) and predicted (purple line) average duration of mechanical ventilation (DMV) over time. The *dashed vertical line* distinguishes the pre-implementation (left of the line) and the post-implementation period (right of the line) as it relates to the ventilator quality improvement initiative. The *red, crosshatched area* indicates the COVID-19 public health emergency (PHE). *Red circles* indicate when the difference between actual and predicted DMV is 5% or greater. The *bottom plot* displays compliance with the ventilator bundle quality measures over time, including spontaneous awakening trials (SATs) (gold line), spontaneous breathing trials (SBTs) (blue line), and low-tidal volume ventilation (LTVV) (red line). A *horizontal dashed line* indicates 90% compliance. The UPMC Model (green line) and electronic Global Open-Source Severity of Illness Score (eGOSSIS) (gray line) standardized mortality ratio (SMR) balancing measures are displayed with values represented on the *right-sided y-axis*.

with the ratio of actual to predicted ventilator days denoted with red dots when it exceeds $\pm 5\%$ of equivalence. **Figure 2** displays the same excluding the potential confounding effect of patients with COVID-19. **Supplemental Figure 19** (<https://links.lww.com/CCM/H734>) displays an extended baseline period. The sums of the predicted classes were less than the sums of the actual classes for all classification DMV models, although there were no significant differences in the distributions of actual to predicted class ratios using the models that did not include length of stay or mortality (**Supplemental Table 16**, <https://links.lww.com/CCM/H734>). A description of the model deployment pipeline is provided in **Supplemental Figure 20** (<https://links.lww.com/CCM/H734>).

DISCUSSION

In our large health system, a multi-hospital QI program focused on a novel dQM bundle coupled with continuous analytics to support iterative audit and feedback was effective at improving process measures related to management of patients requiring mechanical ventilation, with stable to possibly decreased observed vs. predicted ventilator days despite an unadjusted increase in ventilator days. By leveraging an EHR vendor interoperability crosswalk to facilitate data curation, developing care metrics based on the best available evidence, and using robust risk adjustment, our team developed scalable dQMs that were able to efficiently support a QI program across multiple hospitals. This work aligns with the vision of recent federal legislation

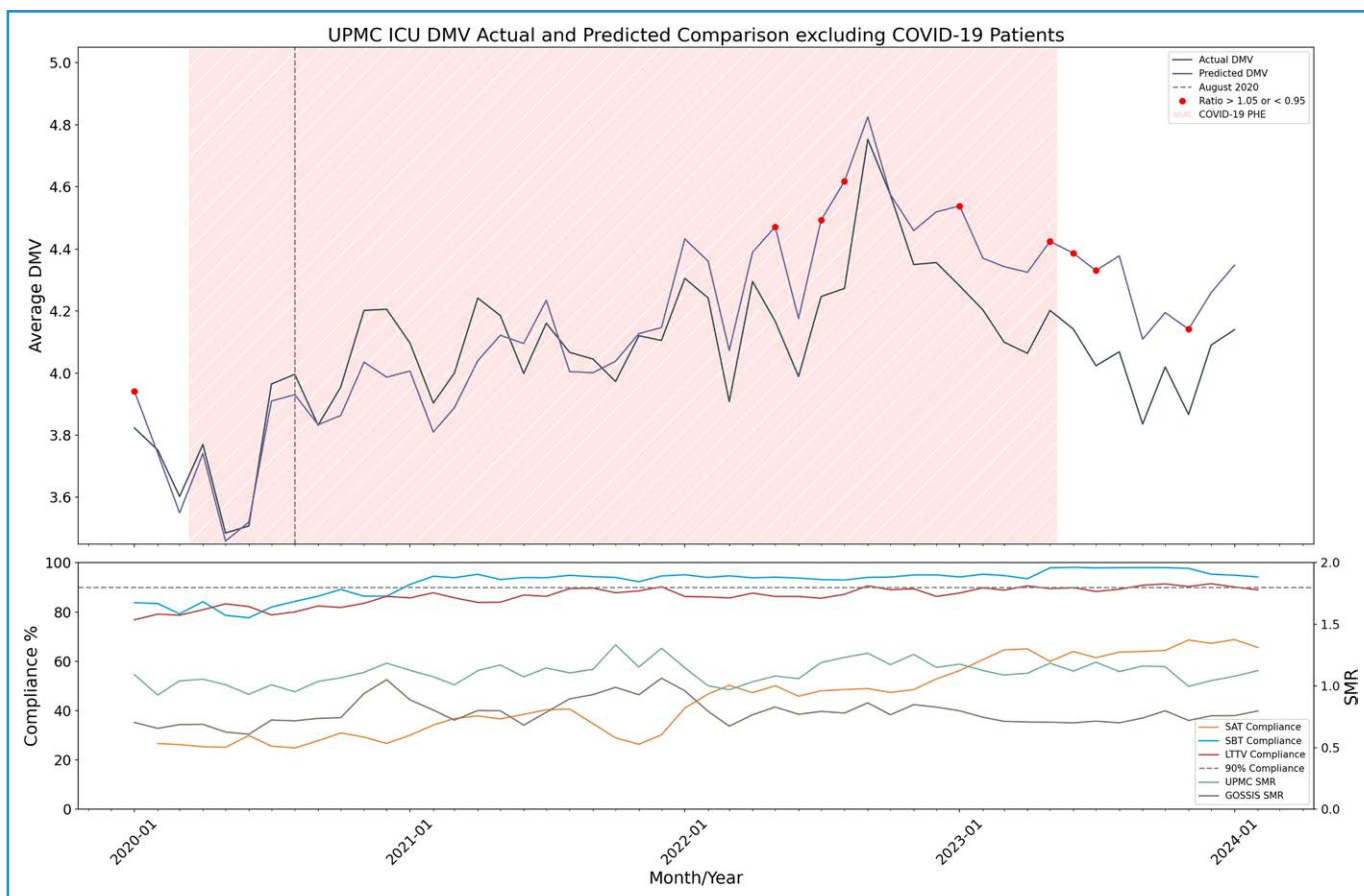


Figure 2. The *top plot* displays actual (black line) and predicted (purple line) average duration of mechanical ventilation (DMV) over time, excluding patients with COVID-19. The *dashed vertical line* distinguishes the pre-implementation (left of the line) and the post-implementation period (right of the line) as it relates to the ventilator quality improvement initiative. The *red, crosshatched area* indicates the COVID-19 public health emergency (PHE). *Red circles* indicate when the difference between actual and predicted DMV is 5% or greater. The *bottom plot* displays compliance with the ventilator bundle quality measures over time, including spontaneous awakening trials (SATs) (gold line), spontaneous breathing trials (SBTs) (blue line), and low-tidal volume ventilation (LTVV) (red line). A *horizontal dashed line* indicates 90% compliance. The UPMC Mortality Model (green line) and electronic Global Open-Source Severity of Illness Score (gray line) standardized mortality ratio (SMR) balancing measures are displayed with values represented on the *right-sided y-axis*.

promoting EHR interoperability and signals the utility of dQMs for improving both patient outcomes and reducing resource utilization.

Although the ventilator dQM bundle was developed for use across our single, large health system, its utility may extend more broadly. The 21st Century Cures Act final rule contains key provisions meant to blaze a path for widespread adoption of electronic QI methods by mandating that EHR vendors in the United States adopt interoperable data standards, including standard vocabularies and transfer standards (16, 27). These steps are meant to help enable the transformation of four domains of quality measurement by: 1) improving data quality, 2) advancing the technology used to support metric tracking, 3) optimizing data aggregation, and 4) aligning data,

tools, and measures (15). Promoting these domains is part of the CMS strategic roadmap for dQMs, which, in a fully actualized form, are expected to lower the overhead associated with reporting QI data (14). dQMs are envisioned to be informed by the best available evidence as an advancement of the National Academy of Medicine's learning health system paradigm (28). By transforming real-world data into information, then knowledge, our work signals the benefits of tracking both care processes and outcomes digitally. However, different information provided by the eGOSSIS vs. Vizient mortality models and 14-day DMV vs. multiple classifier models highlights the importance of generalizable and interpretable model constructs to ensure accurate knowledge leads to appropriate action.

Strengths of the present work include the large patient volume, development of well-performing and up-to-date risk models, the mechanistic plausibility of the association between metrics and actual compared with predicted ventilator days, and the use of best informatics practices to harvest and analyze data. Our approach aligns with the IHI's Model for Improvement framework by regularly engaging ICU leadership through iterative data review and feedback sessions to guide the refinement of practices based on key performance indicators and benchmarked performance data using modern tools such as a digital dashboard (7). The 14 hospitals represented in this work include quaternary, tertiary, and community facilities, suggesting the demonstrated approaches may generalize well across ICUs in the United States. Limitations include the use of a single EHR instance, although UPMC EHR workflows span quaternary, tertiary, and community hospitals suggesting the methods will extend to other institutions. The use of stacked ensemble models obscures feature importance in the duration models. The CMS vision for dQMs includes the use of fast healthcare interoperability resources (FHIRs) and clinical quality language (CQL). Neither FHIR nor CQL were needed for the present work given the single EHR and data retrieval from a local warehouse.

CONCLUSIONS

We developed a dQM bundle that includes robust risk adjustment to promote best ventilator practices across 37 ICUs. An association between improved process and at least stable risk-adjusted mechanical ventilation days was observed. Future work should focus on developing well-calibrated risk adjustment methods that can span disparate health systems, incorporating additional interoperability methods such as FHIR and CQL, and determining whether QI programs relying on interoperable, risk-adjusted dQMs might eventually be associated with improved patient outcomes globally, including lower-resourced health systems with alternative EHRs.

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